Student Performance Prediction System

Project Id: AIML-19

Name : - Divyanshu Panwar

Class Roll Number:45

University Roll Number :2028715

Section and Group: Q, G2

Name : Kunal Joshi

Class Roll Number:51

University Roll Number :2028751

Section and Group: Q, G2

Name : - Nikunj Singhal

Class Roll Number:56

University Roll Number :2028783

Section and Group: Q, G2

1. Problem Description

Air pollution has become a major environmental and public health concern worldwide. Increasing levels of pollutants such as carbon monoxide (CO), nitrogen dioxide (NO_2), and variations in temperature and humidity significantly impact air quality. Poor air quality can lead to serious health issues including respiratory diseases, cardiovascular problems, and reduced life expectancy. Therefore, there is a need for an intelligent system that can accurately classify air quality levels based on sensor data.

This project aims to design and develop a machine learning-based Air Quality Level Classification System that predicts air quality categories (e.g., Good, Moderate, Poor) using environmental and pollutant data. The system will utilize features such as CO, NO_2, temperature, and humidity collected from sensors. Data preprocessing techniques like handling missing values, noise removal, and feature scaling will be applied to improve data quality. Feature selection methods will help in identifying the most relevant parameters affecting air quality.

Classification algorithms such as K-Nearest Neighbors (KNN) and Logistic Regression will be used to build predictive models. The performance of these models will be evaluated using metrics like accuracy and precision. This system will help in monitoring air pollution levels and assist authorities and individuals in making informed decisions.

2. Architecture

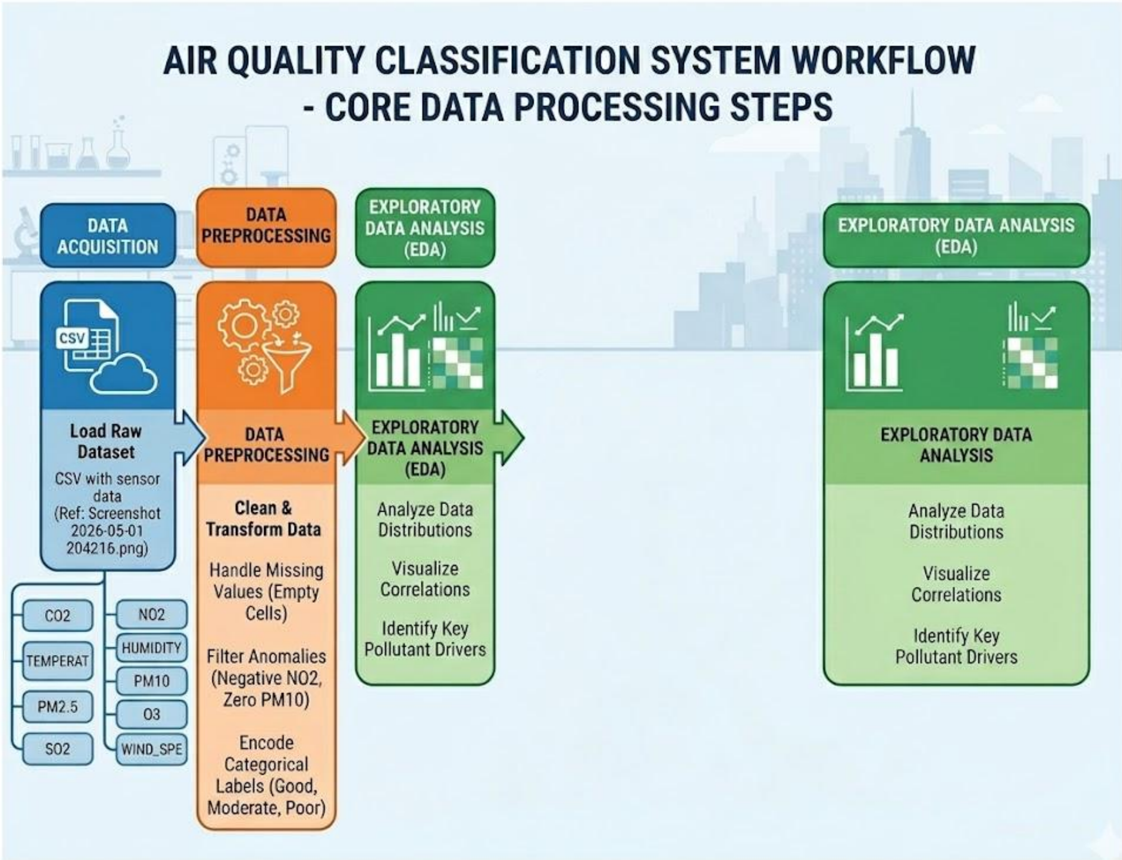


Fig.1 Sample flow diagram.

3. Models Used(Applied/ Planned) Give a brief introduction of the models used

In this project, machine learning classification models are used to predict air quality levels based on environmental and pollutant data such as CO, NO_2, temperature, humidity. The following models are applied/planned:

1. K-Nearest Neighbors (KNN)

    K-Nearest Neighbors (KNN) is a supervised machine learning algorithm used for classification tasks. It works by identifying the 'k' closest data points (neighbors) to a new input sample based on distance metrics such as Euclidean distance. The class label is then assigned based on the majority class among these neighbors.

    KNN is simple, effective, and works well with smaller datasets. It does not require a training phase, making it easy to implement. However, its performance depends on the choice of 'k' and can be sensitive to noisy data.

2. Logistic Regression

    Logistic Regression is a statistical classification algorithm used to predict categorical outcomes. It estimates the probability that a given input belongs to a particular class using a logistic (sigmoid) function.

    Despite its name, Logistic Regression is widely used for classification problems rather than regression. It is efficient, interpretable, and performs well when the relationship between features and the target variable is linear. It also provides probability outputs, which help in understanding prediction confidence.

4. Evaluation metrics used

To assess the performance of the machine learning models used for air quality classification, several evaluation metrics are employed. These metrics help in understanding how well the models predict air quality categories such as Good, Moderate, and Poor.

1. Accuracy Score
Accuracy is calculated using the accuracy_score function. It measures the overall correctness of the model by comparing predicted labels with actual labels. It is one of the most commonly used metrics for classification problems.

2. Precision Score
Precision is computed using precision_score. It indicates how many of the predicted positive cases (e.g., Poor air quality) are actually correct. This is useful when false alarms need to be minimized.

3. Recall Score
Recall is calculated using recall_score. It measures how well the model identifies actual positive cases. This is important when missing critical cases (like hazardous air quality) should be avoided.

4. F1 Score
F1 Score is obtained using f1_score. It provides a balance between precision and recall, making it useful when both false positives and false negatives are important.

5. Confusion Matrix
The confusion matrix is generated using confusion_matrix. It provides a detailed breakdown of correct and incorrect predictions across all classes, helping to better understand model performance.

5. Preliminary Results and Visualizations

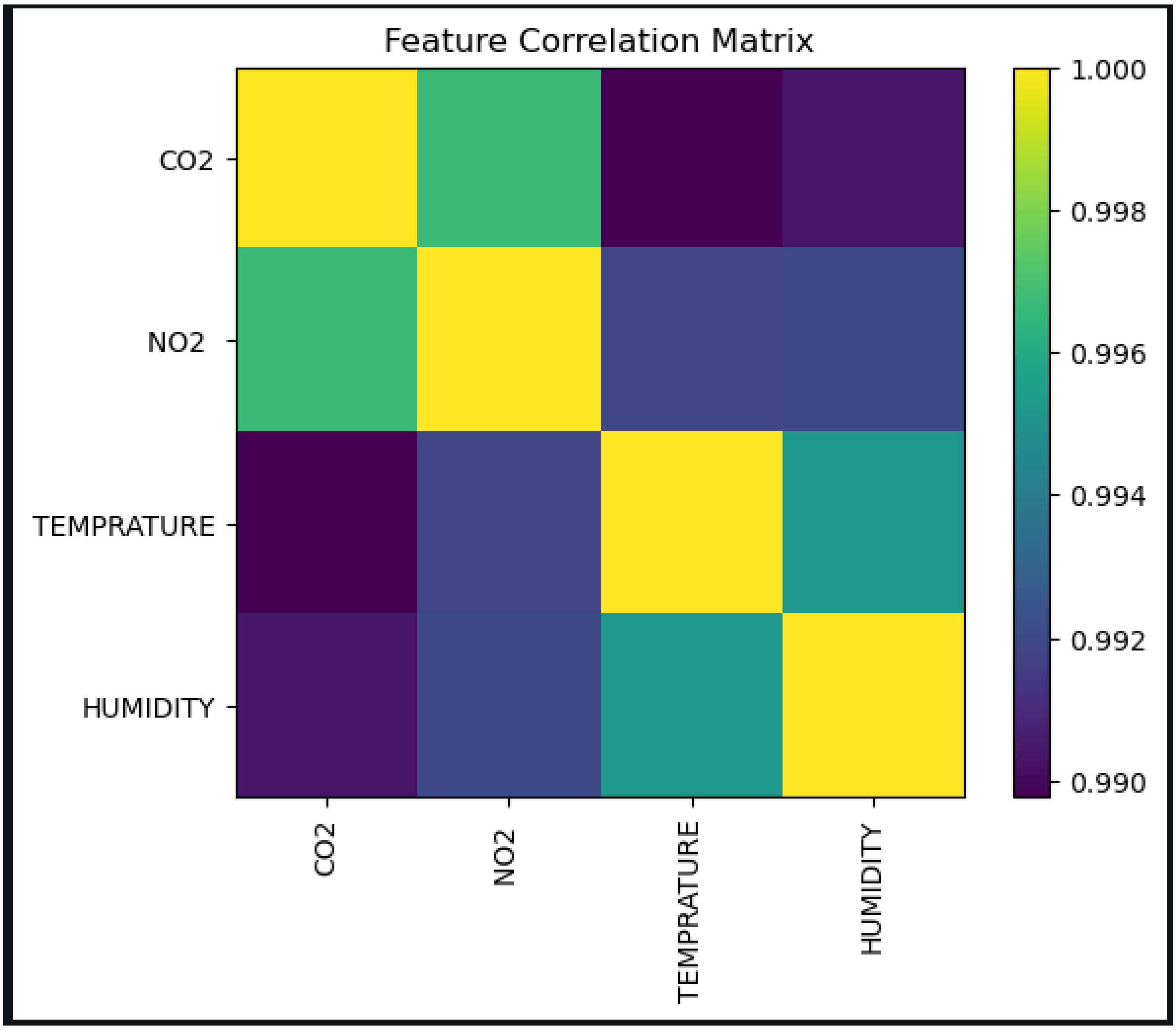
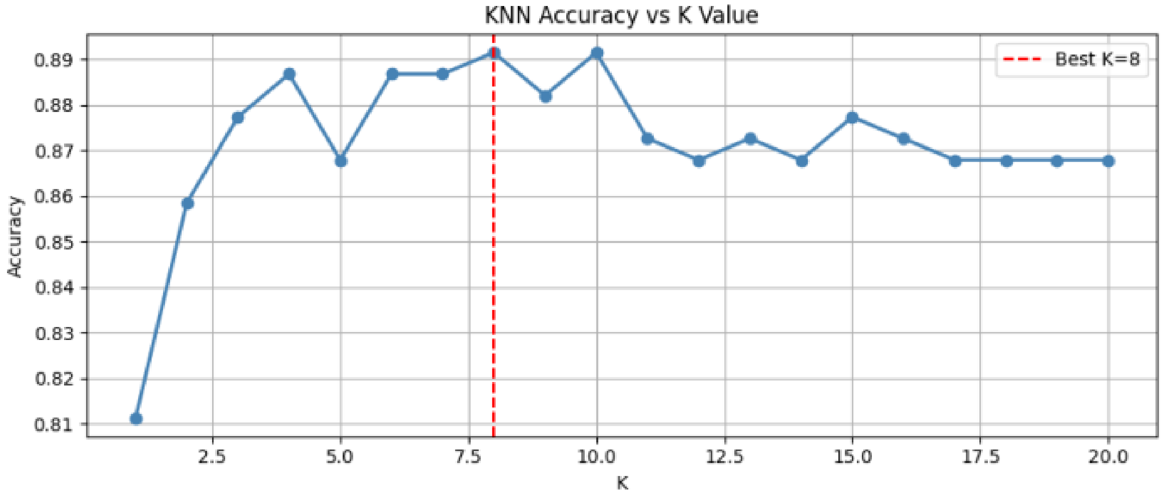


Fig.2 Heatmap showing corelation.


KNN Accuracy vs K Value
Best K=8
0.89
0.88
0.87
0.86
0.85
0.84
0.83
0.82
0.81
Accuracy
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0
K
Fig.3

6. References:

Books:
1. Let Us Python – Yashavant Kanetkar

Online Resources:
• GeeksforGeeks – https://www.geeksforgeeks.org/
• TutorialsPoint – https://www.tutorialspoint.com/cprogramming/index.htm
• Stack Overflow – https://stackoverflow.com
• Pandas Documentation – For data preprocessing and dataset handling.
• NumPy Documentation – For numerical operations and array handling